

Bitcoin Miner - Maze Collection Game

A fun top-down maze game where you play as a Bitcoin miner collecting coins while avoiding ghosts! Collect all coins, then deposit them in the wallet to win.

Game Description

Navigate through a maze, collect all the floating Bitcoin and Dollar coins, then reach the wallet to deposit them and win! But watch out — a ghost is chasing you, and hitting walls will send you back to the start.

Game Rules

- **Collect** all 10 coins scattered throughout the maze
 - **Avoid** the ghost — if it catches you, you respawn at the start
 - **Don't hit walls** — they also send you back to the start
 - **Touch the wallet** after collecting all coins to win!
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Features

- Smooth player movement with arrow keys or WASD
 - Smart ghost AI that chases you around walls
 - Floating coin animations (Bitcoin & Dollar)
 - Glowing wallet that activates when all coins are collected
 - Particle effects when collecting coins
 - Sound effects for coin collection, win, and lose
 - Background music
 - Play again after winning or losing
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Prerequisites

Before you start, you need:

1. **Python 3.7 or higher**
 2. **Pygame library**
 3. **Game assets** (images and sounds)
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Windows Setup

Step 1: Install Python

1. Go to python.org/downloads
2. Download the latest Python 3 installer
3. **IMPORTANT:** During installation, check the box that says “**Add Python to PATH**”
4. Click “Install Now”

5. Open Command Prompt and verify:

```
python --version
```

You should see something like Python 3.11.x

Step 2: Install Pygame

Open Command Prompt and run:

```
pip install pygame
```

If that doesn't work, try:

```
python -m pip install pygame
```

Step 3: Download the Game

1. Create a new folder for the game:

```
mkdir C:\Users\%USERNAME%\bitcoin_miner  
cd C:\Users\%USERNAME%\bitcoin_miner
```

2. Download bitcoin_miner_maze.py and place it in this folder

3. Download all game assets and place them in the **same folder**:

- background.jpg
- miner.png
- bitcoin.png
- dollar.png
- ghost.png
- cave.mp3
- coin_drop.mp3
- win.wav
- lose.wav

Step 4: Run the Game

In Command Prompt, run:

```
cd C:\Users\%USERNAME%\bitcoin_miner  
python bitcoin_miner_maze.py
```

Or simply double-click the bitcoin_miner_maze.py file!

Linux Setup (Fedora / Ubuntu / Debian)

Step 1: Install Python

Most Linux distributions come with Python pre-installed. Check if you have it:

```
python3 --version
```

If not installed, install it:

Fedora:

```
sudo dnf install python3 python3-pip
```

Ubuntu/Debian:

```
sudo apt update  
sudo apt install python3 python3-pip
```

Arch:

```
sudo pacman -S python python-pip
```

Step 2: Install Pygame

```
pip3 install pygame --user
```

If you get a “externally-managed-environment” error, try:

```
pip3 install pygame --user --break-system-packages
```

Or use your package manager:

Fedora:

```
sudo dnf install python3-pygame
```

Ubuntu/Debian:

```
sudo apt install python3-pygame
```

Arch:

```
sudo pacman -S python-pygame
```

Step 3: Download the Game

Create a folder and download everything:

```
mkdir -p ~/bitcoin_miner  
cd ~/bitcoin_miner
```

Place these files in the folder:

- bitcoin_miner_maze.py (the game code)
- background.jpg
- miner.png
- bitcoin.png
- dollar.png
- ghost.png

- cave.mp3
- coin_drop.mp3
- win.wav
- lose.wav

Step 4: Run the Game

```
cd ~/bitcoin_miner
python3 bitcoin_miner_maze.py
```

How to Play

1. **Start the game** — you'll see the title screen
2. **Press SPACE or CLICK** to begin
3. **Move around** using arrow keys or WASD
4. **Collect all coins** — they float up and down
5. **Watch the ghost** — it will chase you!
6. **Avoid walls** — hitting them sends you back to start
7. **When all coins are collected**, the wallet starts glowing gold
8. **Touch the wallet** to deposit your coins and WIN!

Controls

Key	Action
UP / W	Move Up
DOWN / S	Move Down
LEFT / A	Move Left
RIGHT / D	Move Right
SPACE	Start Game / Play Again
ESC	Quit Game
Mouse Click	Start Game / Play Again

Troubleshooting

Problem: “No module named ‘pygame’”

Solution: Pygame is not installed. Install it:

- Windows: `pip install pygame`
- Linux: `pip3 install pygame --user`

Problem: Pink/Magenta squares instead of images

Solution: The game can't find your image files. Make sure all .png and .jpg files are in the **same folder** as bitcoin_miner_maze.py.

Problem: No sound

Solution: Check that .mp3 and .wav files are in the game folder. Also check your system volume!

Problem: Game runs too fast or too slow

Solution: The game is locked to 60 FPS. If it's still too fast, your monitor might have a high refresh rate.

Problem: "Permission denied" on Linux

Solution: Make the file executable:

```
chmod +x bitcoin_miner_maze.py
```

Problem: Ghost is too hard / too easy

Solution: Open bitcoin_miner_maze.py in a text editor and find this line in the Ghost class:

```
self.speed = 2.5
```

Change the number to make it slower (e.g., 1.5) or faster (e.g., 4.0).

File Structure

Your game folder should look like this:

```
bitcoin_miner/
├─ bitcoin_miner_maze.py    # Main game file (the code)
├─ background.jpg          # Cave background image
├─ miner.png               # Player character
├─ bitcoin.png             # Bitcoin coin collectible
├─ dollar.png              # Dollar coin collectible
├─ ghost.png               # Enemy ghost
├─ cave.mp3                # Background music
├─ coin_drop.mp3           # Coin collection sound
├─ win.wav                 # Victory sound
├─ lose.wav                # Game over sound
└─ README.md               # This file!
```

Customization

Want to change things? Here are some ideas:

Change the number of coins

Find this line in the code:

```
coins = spawn_coins(walls, 10)
```

Change **10** to any number you want!

Change player speed

Find this line in the Player class:

```
self.speed = 4
```

Make it bigger to go faster, smaller to go slower.

Change maze layout

Look for `generate_walls()` function and modify the wall coordinates!

License

This game is for educational and personal use. Feel free to modify and share!

Credits

- Built with Pygame
 - Inspired by classic arcade games
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Need Help?

If you're stuck:

1. Make sure all files are in the same folder
2. Check that Python and Pygame are installed correctly
3. Read the error message carefully — it usually tells you what's wrong!

Happy Mining!